Exercises Week 1

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Exercise 1

Let $X(t)$ be a Gaussian zero-mean process with t $\in \mathbf{R}^d$, d = 1,2, and with Matérn covariance function.

- Given the full Matérn covariance, let us construct a discrete approximation $\mathbf{C}$, and then compute the modified Cholesky factorisation $\mathbf{C} = \mathbf{U}^T + \mathbf{U}$ with discretisation
  - t = ih, h = 0.01, i = 0,...,100 in dimension one

$$t_i \in [0, 1] \tag{1}$$

The distance D_{ij}

$$D_{ij} = ||t_i - t_j||_2 \tag{2}$$

The Matérn covariance

$$\begin{aligned} C_v(d_{ij}) &= \sigma^2 \frac{2^{1-v}}{\Gamma(v)} \left(\frac{\sqrt{2v}d}{l} \right)^v K_v \left(\frac{\sqrt{2v}d}{l} \right), i \neq j \\ C_v(d_{ii}) &= \sigma_f^2, \forall i \in \{1, ..., n\} \end{aligned} \tag{3}$$

(Rasmussen, 2005)
The U matrix for Cholesky

$$\begin{aligned} U_{00} &= \sqrt{C_{00}} = \sigma_f \\ U_{ii} &= \sqrt{C_{ii} - \sum_{k=0}^{i-1} U_{ki}^2}, \forall i \in \{1, .., 100\} \\ U_{0j} &= \frac{C_{0j}}{U_{00}} \\ U_{ij} &= \frac{1}{U_{ii}} \left(C_{ij} - \sum_{k=0}^{i-1} U_{ki} U_{kj} \right), \text{for } j > i \end{aligned} \tag{4}$$

(Pourahmadi, 2014)

$$W_w \sim N(0, 1) \tag{5}$$

$$X_w = UW_w \tag{6}$$

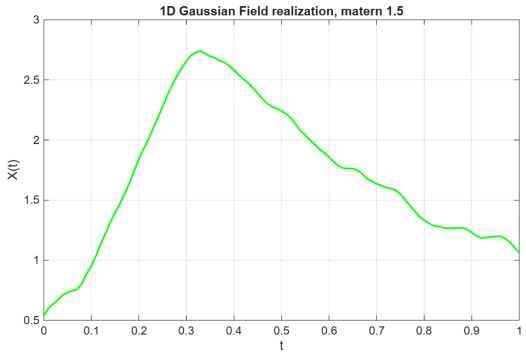


Figure 1: 1D Gaussian Field Realization

■ t $= (h_1, h_2)^T$ on equidistant lattice in dimension two with same lattice as for the one-dimensional case.

Let us denote our grid points as (x_m, y_n) where m$\in \{0, 1...N_x - 1\}$ and n$\in \{0, 1...N_y - 1\}$. So $N = N_x + Ny$.
Let us map each grid node to the index i $\in \{1, 2..., N\}$
$i(m, n) = m \cdot N_y + n + 1$

Then t_i is expressed as $t_i = (x_i, y_i)^T = (m_i h, n_i h)^T = ([(i-1)/N_y]h, [(i-1) \mod N_y]h)^T$
The pairwise distance in the grid is given by

$$\begin{aligned} D_{ij} &= h \cdot \sqrt{(m_i - m_j)^2 + (n_i - n_j)^2} \\ &= h \cdot \sqrt{([(i-1)/N_y] - [(j-1)/N_y])^2 + (((i-1) \mod N_y) - ((j-1) \mod N_y))^2} \end{aligned} \tag{7}$$

Now the 2D Matérn covariance is

$$\begin{aligned} C_{ii} &= \sigma_f^2 \\ C_{ij} &= \sigma_f^2 \cdot \frac{2^{1-v}}{\Gamma(v)} \left(\frac{\sqrt{2v} D_{ij}}{l} \right)^v K_v \left(\frac{\sqrt{2v} D_{ij}}{l} \right), \text{if i} \neq \text{j} \end{aligned} \tag{8}$$

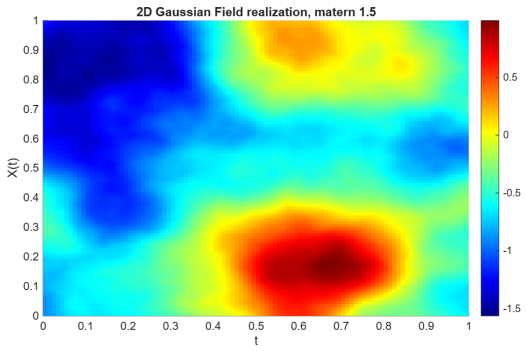


Figure 2: Caption

Exercise 2

pCn for GP regression: let us simulate a Gaussian process regression for this case. In order to sample the states, we will use preconditioned Crank-Nicolson.

- One realization X_w of the one-dimensaional Matérn process as in T1

$$W_w \sim N(0,1) \tag{9}$$
$$X_w = U^T W_w \tag{10}$$

- GP regression model using closed-form matrix formulas to obtain the conditional mean and posterior covariances. The changes in number of measurements and states can be seen affecting the posterior mean and covariance estimates.

  Closed form GPR
  we have noisy observations $y = Hx + \epsilon$ observed at indices Idx_{obs}. The errors follow $\epsilon \sim N(0, \sigma_n^2 I_K)$

$$K_{obs} = C(Idx_{obs}, Idx_{obs} \tag{11}$$
$$K_c = C(:; Idx_{obs}) \tag{12}$$

  We can calculate the analytical posterior mean and covariance

$$\mu_{post} = K_c K_{obs}^{-1} y \tag{13}$$
$$\Sigma_{post} = C - K_c K_{obs}^{-1} K_c^T \tag{14}$$

  (Rasmussen, 2005)
  With the potential function $\Phi(x)$, we can denote the likelihood $p(y|x)$ as

$$p(y|x) \propto \exp\left(-\Phi(x)\right) \tag{15}$$

  so that $\Phi(x)$ is defined as

$$\Phi(x) = \frac{1}{2\sigma_n^2} \sum_{k=1}^{K} (y_k - x(Idx_{obs}(k)))^2 \tag{16}$$

  (Calvetti, 2023)

- Instead of direct GP regression, let us employ MCMC methods for state estimation. Specifically both Metropolis-Hastings and preconditioned Crank-Nicolson are implemented.

  The proposal step generates states updates based on the prior distribution.

$$x^{(k+1)} = \sqrt{1-\beta^2} x^{(k)} + \beta U^T z, z \sim N(0, I_N) \tag{17}$$

  (Beskos, 2014)
  Now the acceptance probability is based on the difference of likelihood potential.

$$\alpha = \min(1, \exp(\Phi(x^{(k)}) - \Phi(x^{(k+1)}))) \tag{18}$$

  The acceptance rate of pCN MCMC is 34% and 0% for Metropolis Hastings when burn-in is 500.

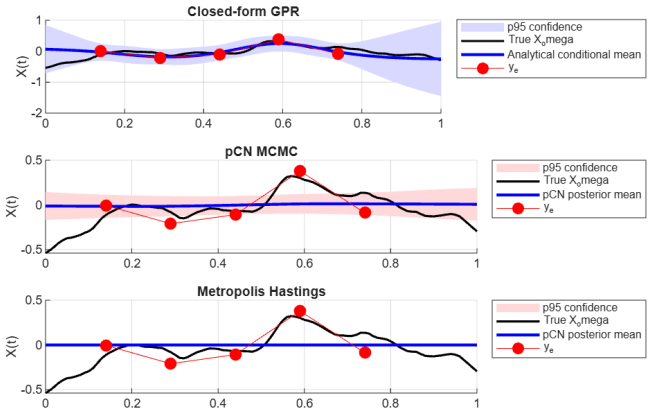


Figure 3: State estimation of 1D Matérn closed-form GPR, pCN MCMC and Metropolis Hastings

Parameters

N	101
N_{mcmc}	1000
burnIn	500
u_0	0_{101*1}
v	1.5
σ_f^2	1.0
l	0.2
jitter	$10^{-8}I$
observation locations	$15\%, 30\%, 45\%, 60\%, 75\%$
σ_n	0.1
β	0.15
δ	0.03

References

Beskos, Alexandros, R. G. S. A. (2014). MCMC Methods for diffusion bridges. *Stochastics and Dynamics*, 8(2):561–586.

Calvetti, Daniel, S. E. (2023). *Bayesian Scientific Computing*. Cham: Springer.

Pourahmadi, M. (2014). *High-Dimensional Covariance Estimation: With High-Dimensional Data*. Newark: John Wiley et Sons Ltd.

Rasmussen, Carl Edward, W. C. K. (2005). *Gaussian Processes for Machine Learning*. MIT Press.